Standard Operating Procedure for Resolving Hardware/Environmental Faults in Ericsson PE (IND-PE-03) at Indiranagar Exchange

Node: IND-PE-03 | **Vendor:** Ericsson | **Severity:** Warning | **Alarm Code:** ALARM_6508

Description

The failure domain of hardware/environmental faults in the PE (IND-PE-03) at Indiranagar Exchange is primarily attributed to the inherent susceptibility of the ASIC memory parity errors and line card crashes, compounded by the loss of power supply redundancy. This vulnerability is rooted in the underlying protocol mechanics of the Ericsson PE, where the Line Card Interface (LCI) manages the interactions between the Line Card and the Control Card, facilitating the exchange of control and signaling information. The LCI relies on a redundant power supply configuration to ensure uninterrupted operation. However, when the power supply redundancy is lost, the LCI may experience memory parity errors, leading to line card crashes and subsequent alarm generation. Additionally, environmental factors such as temperature fluctuations, humidity, and contamination may also contribute to hardware malfunctions, exacerbating the fault domain. This SOP aims to provide a comprehensive approach to diagnosing and resolving hardware/environmental faults in the Ericsson PE at Indiranagar Exchange, ensuring minimal downtime and maximizing network availability.

Pre-requisite Checks

- Dispatch remote hands to Indiranagar Exchange for physical access and assistance

- Verify the redundant link status and ensure that the link is up and operational

- Check the power supply redundancy configuration to confirm that the primary and secondary power supplies are functioning correctly

- Review the Ericsson PE logs to identify any recent errors or anomalies related to the fault

Potential Root Causes

Cause	Probability
ASIC memory parity errors	High
Line card crashes	Medium
Loss of power supply redundancy	Low

Diagnostic Commands & Expected Outputs

```
Command: show diag lc [all] | grep 'Error', Expected Output: Error:
Parity error detected on LCI [X], Description: This command will
display the diagnostic information for all line cards (or the
specified line card) and check for any parity errors. The 'grep'
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```
command is used to filter the output and display only the lines
containing the string 'Error'. This will help identify any parity
errors that may have occurred on the line card.

Command: show diag lci [all] | grep 'Power Supply', Expected Output:
Power Supply Status: Down, Description: This command will display the
diagnostic information for all LCI (or the specified LCI) and check
the power supply status. The 'grep' command is used to filter the
output and display only the lines containing the string 'Power
Supply'. This will help identify any issues related to the power
supply redundancy.

Command: show env [all] | grep 'Temperature', Expected Output:
Temperature: 45°C (out of range), Description: This command will
display the environmental information for all (or the specified) line
cards and check the temperature. The 'grep' command is used to filter
the output and display only the lines containing the string
'Temperature'. This will help identify any environmental issues that
may be contributing to the fault.
```

Resolution Steps

Step 1: Step: Swap the line card with a spare, Command: replace lc [X] with [spare-X], Description: This command will swap the faulty line card with a spare line card. The 'replace' command is used to replace the line card, and the '[spare-X]' parameter specifies the spare line card to be used.

Step 2: Step: Check and restore power supply redundancy, Command: show diag lci [all] | grep 'Power Supply', Description: This command will display the diagnostic information for all LCI (or the specified LCI) and check the power supply status. If the power supply status is found to be down, the command to restore power supply redundancy will be executed.

Step 3: Step: Review and adjust environmental settings, Command: show env [all] | grep 'Temperature', Description: This command will display the environmental information for all (or the specified) line cards and check the temperature. If the temperature is found to be out of range, the environmental settings will be reviewed and adjusted to ensure optimal operating conditions.

Post-Check Verification

```
Command: show diag lc [all] | grep 'Error', Expected Output: Error:
None, Description: This command will display the diagnostic
information for all line cards (or the specified line card) and check
for any parity errors. The 'grep' command is used to filter the output
and display only the lines containing the string 'Error'. This will
help verify that the parity errors have been resolved.

Command: show diag lci [all] | grep 'Power Supply', Expected Output:
Power Supply Status: Up, Description: This command will display the
diagnostic information for all LCI (or the specified LCI) and check
the power supply status. The 'grep' command is used to filter the
output and display only the lines containing the string 'Power
Supply'. This will help verify that the power supply redundancy has
been restored.

Command: show env [all] | grep 'Temperature', Expected Output:
Temperature: within range, Description: This command will display the
environmental information for all (or the specified) line cards and
check the temperature. The 'grep' command is used to filter the output
and display only the lines containing the string 'Temperature'. This
will help verify that the environmental settings have been adjusted to
optimal operating conditions.
```

Roll-back Procedures

- Step: Restore the original line card, Command: replace lc [X] with [original-X], Description: This command will restore the original line card that was swapped out. The 'replace' command is used to replace the line card, and the '[original-X]' parameter specifies the original line card to be restored.

- Step: Restore power supply redundancy settings, Command: config lci [X] power-supply [primary/secondary], Description: This command will restore the power supply redundancy settings to their original configuration. The 'config' command is used to configure the LCI, and the '[primary/secondary]' parameter specifies the power supply to be restored.

- Step: Restore environmental settings, Command: config env [X] temperature [value], Description: This command will restore the environmental settings to their original configuration. The 'config' command is used to configure the environment, and the '[value]' parameter specifies the temperature setting to be restored.